

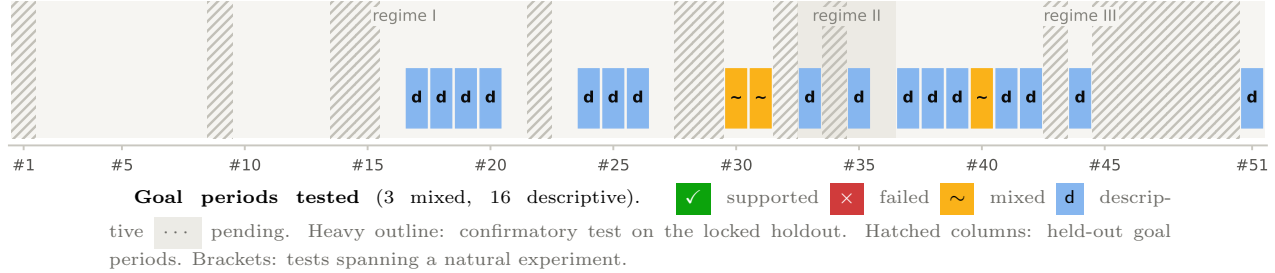
H27: Critical slowing down warns of herding waves hours ahead

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

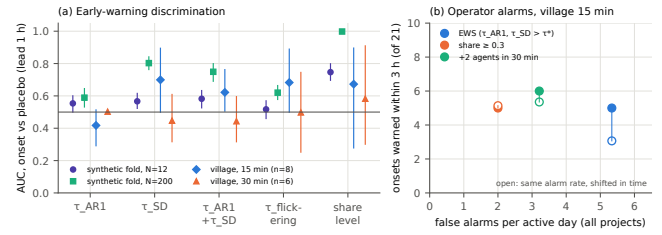
Bottom line. Critical slowing down does not warn of village herding: pile-ons in attention and commits often follow a link within minutes, and $N \approx 12$ hides slowing.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

A pre-registered rule on project labels defines a herding onset. A project's share of labeled agents jumps from ≤ 0.25 to ≥ 0.5 within an hour and stays there. It finds 21 onsets in 14 non-holdout weeks at 15-min resolution, mostly in shared-artifact weeks. We compare early-warning trends (rolling lag-1 autocorrelation and SD over the previous 6 h) with matched placebo segments; 8 onsets are evaluable. The composite separates them weakly at 15 min (AUC 0.62) and not at 30 min. Autocorrelation never rises (AUC 0.41), so this is not critical slowing down. **Round 1b** (deterministic labels, work ledger) reproduces all of it: the same onsets and the same AUC. The frozen alarm now beats a time-shifted copy at $p = 0.038$ (was 0.07). A naive share alarm matches its 5/21 hits with fewer alarms. Work-commit onsets coincide with attention onsets. The #40 pile-on ran in minutes after a link. The holdout is not run.



Observables. Round 1. (a) AUC of each early-warning trend, onset vs placebo segments, at 1 h lead. Series: a synthetic fold at village size (purple), a clean $N = 200$ fold (green), and the village at 15 min (blue, 8 onsets) and 30 min (orange, 6). Autocorrelation never discriminates. Variance does only at 15 min, with wide CIs. (b) Onsets warned within 3 h vs false alarms per active day. Open circles: the same alarms shifted in time. In round 1, no alarm beats its shifted copy.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not build a critical-slowness dashboard for a 10–15-agent LLM swarm: rising autocorrelation and variance do not forecast its pile-ons. A pile-on is a fast push: a link in chat, then half the room within minutes. In #40, half the room's attention took 4 min and half its commits 19 min. A 6-hour trend window cannot see that, and at $N \approx 12$ a true fold is barely detectable. Watch announcements and project shares instead, and plan responses in minutes, not hours. Test any alarm against a rate-matched time shift: here p moves from 0.07 to 0.038 between data versions.

PHYSICAL MODEL

Kinetic mean-field Potts (10) with mean-field susceptibility (01). Spin σ_i : the project of agent i in a window (0 = rare projects, uncoupled). $k_a^{(-i)}$: other agents on a . $h_a(t)$: a slowly drifting popularity field. βJ : uniform herding coupling. γ : reconsideration rate. Order parameter x_a : the share on a . Near a fold of x_A the leading eigenvalue $\lambda \rightarrow 0$, so the lag-1 autocorrelation ρ_1 and the variance rise (critical slowing down). Rivals (field steps, noise-induced flips) give no warning.

$$P(\sigma_i \rightarrow a) \propto e^{h_a(t) + \beta J k_a^{(-i)} / (N-1)} \quad (a \geq 1), \quad \propto 1 \quad (a = 0)$$

$$\dot{x}_a = \gamma [\text{softmax}_a(h(t) + \beta J x) - x_a]$$

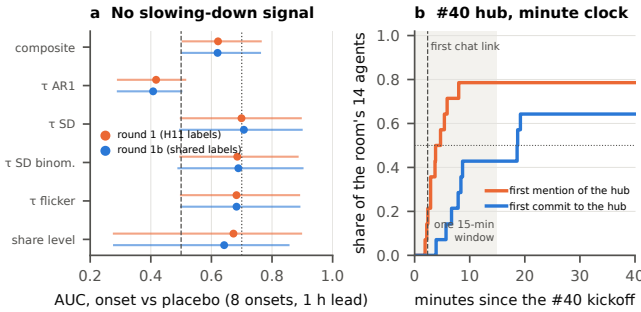
$$\rho_1 = e^{-\gamma |\lambda| \Delta t} \rightarrow 1, \quad \text{Var } x_A \propto 1/(N|\lambda|) \quad (\lambda \rightarrow 0)$$

$$\text{alarm: } \tau_K(\hat{\rho}_1) > \tau^* \wedge \tau_K(\hat{\sigma}) > \tau^*, \quad \tau^* = 0.54$$

H27-herding-early-warning · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H27-herding-early-warning/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0)

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P0: 10–25 onsets	21 → 21; 8 evaluable	yes
P1: $AUC \geq 0.70$	0.62 → 0.62 [0.50, 0.76]; 30 min 0.44 → 0.50	mixed
P2: $SD > AR1$	0.70/0.42 → 0.71/0.41	partly
P3: no operator value	5/21 hits; shift p 0.07 → 0.038; level alarm 5/21	mostly
P4: synthetic	$N=200$: 0.75; $N=12$: 0.58	2 of 4
R1b-1 replication	same onsets, same verdicts	yes
R1b-2 work on-sets	8 vs 14; lag 0 (3) or +1 (1) window	mostly
R1b-3 links	11/21; 9 to known projects, 2 first links	yes
#40 minute clock	t_{50} 1.5 min after link; work 15 min later	2 of 3



Second figure. (a) AUC of each trend, onset vs placebo, at 1 h lead: round 1 (orange) vs round 1b shared labels (blue). Nothing moves; autocorrelation stays below 0.5. (b) #40 at the minute clock: room share with a first hub mention (orange) or first hub commit (blue) after the kickoff. Half the room joins within one 15-min window (shaded).

GOAL PERIODS IN DETAIL

G17 d 1b: descriptive (unchanged)
 G18 d 1b: descriptive (unchanged)
 G19 d 1b: descriptive (unchanged)
 G20 d 1b: descriptive (unchanged)
 G24 d 1b: descriptive (unchanged)
 G25 d 1b: descriptive (unchanged)
 G26 d 1b: descriptive (unchanged)
 G30 ~ 1b: mixed (AUC 0.55, 2/4 hit; unchanged)
 G31 ~ 1b: mixed (native: work onsets coincide with attention)
 G33 d 1b: descriptive (unchanged)
 +9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

There are 21 onsets, 8 evaluable and 4 in work, so per-week AUCs rest on 1–2 onsets. We tried many indicators, leads, window lengths and two onset rules; only the 15-min composite at 1 h was primary. The alarm-vs-shift p crosses 0.05 between null draws (0.038–0.07): treat it as borderline. The step rule misses slow rises; the slow-rise variant gives 0.74 at 15 min. The link association is post hoc, and most linked onsets are bursts of known projects (H28). Work labels are public agent commits (dense from #30), and #51 supplies most false alarms.

ROUND 2

- **Essence.** Can an operator see a pile-on coming early enough to act, and from what signal?
- **H27-R1** Model onsets as announcement-triggered cascades: hazard of a majority pile-on after a link post, by poster, room size and the project's prior share; joint with H28.
- **H27-R2** Use a finer clock: per-turn project states from actions (minutes, not 15-min windows) so 6 h of history is not needed and the first 1.5 days become evaluable.
- **H27-R3** Test the one regime where the theory says warnings can work: long, large swarms with slowly drifting incentives (#51 roles, if any shared project ever rises), with the slow-rise onset rule.